LIRA — Network Security Investigation Report

> **ML detects. LLM investigates, explains and guides.**

1. Executive Summary

- **Investigation ID:** `8f62546e-fa23-4116-a5d8-215c48964b0d`
- **Status:** COMPLETED
- **Predicted traffic class:** `BENIGN`
- **Random Forest confidence:** 100.0%
- **Additional analyses used:** 3
- **Analyses remaining:** 0
- **Report generated:** 2026-11-02 12:20 UTC

Preliminary assessment

The model classified this flow as **BENIGN**. This is a model prediction, not independent proof that a real-world attack occurred.

2. Investigation History

- Reason: Prediction analysis shows BENIGN at 1.0 confidence with LOW uncertainty and data quality is VALID, but dataset-label context and cautions have not yet been reviewed. Need to clarify what BENIGN prediction can and cannot prove before forming a provisional conclusion.

3. Investigation Evidence

Analysis 1 — `analyze_prediction`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-02T12:20:00.307456+00:00
- **Reason:** Initial detection is BENIGN with 1.0 confidence but no evidence has been collected; need to assess prediction ambiguity and uncertainty before concluding.


```
{
  "predicted_class": "BENIGN",
  "status": "NORMAL",
  "model_confidence": 1.0,
  "top_probability": 1.0,
  "second_best_class": "DDoS",
  "probability_gap": 1.0,
  "prediction_uncertainty": "LOW",
  "requires_further_investigation": false,
  "interpretation": "The leading class is clearly ahead of alternatives.",
  "limitations": [
    "Random Forest probabilities may not be calibrated.",
    "A high probability does not prove an attack occurred."
  ]
}
```


Analysis 2 — `check_data_quality`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-02T12:20:05.056135+00:00
- **Reason:** BENIGN prediction at 1.0 confidence is not proof; data quality has not been verified and is required to determine if input features are usable before concluding.


```
{
```

```
  "tool": "check_data_quality",
  "valid": true,
  "status": "VALID",
  "rows_checked": 1,
  "features_checked": 78,
  "missing_or_non_numeric_values": 0,
  "infinite_values": 0
}
```


Analysis 3 — `review_class_context`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-02T12:20:10.366576+00:00
- **Reason:** Prediction analysis shows BENIGN at 1.0 confidence with LOW uncertainty and data quality is VALID, but dataset-label context and cautions have not yet been reviewed. Need to clarify what BENIGN prediction can and cannot prove before forming a provisional conclusion.


```
{
  "tool": "review_class_context",
  "status": "COMPLETED",
  "predicted_class": "BENIGN",
  "dataset_description": "Traffic labeled as normal in the reference dataset.",
  "caution": "A BENIGN prediction does not guarantee traffic is safe.",
  "scope": "This describes dataset labels, not verified live incidents."
}
```


4. Final Assessment

Provisional conclusion: Model predicts BENIGN with 1.0 confidence (probability gap 1.0, LOW uncertainty) and data quality is VALID (1 row, 78 features, no missing/infinite values). Dataset context indicates BENIGN means 'normal' in reference labels, but this does not guarantee live traffic is safe. Limitations remain: Random Forest probabilities may be uncalibrated and a prediction is not proof of real-world status. No evidence of attack was presented, but safety cannot be conclusively verified from prediction alone; continued monitoring is warranted.

5. Recommended Operational Follow-up

1. Check relevant server, authentication, and network logs for corroborating evidence.
2. Verify whether the activity matches legitimate administrative or testing activity.
3. Review the alert with a network or security administrator before taking disruptive action.
4. Do not treat model confidence or similarity to reference flows as definitive proof of compromise.

6. Limitations

- The investigation concerns a dataset record, not live production traffic.
- Random Forest probabilities may not be calibrated.
- Similarity to reference flows does not independently confirm an attack.
- No independent operational logs are available to confirm compromise.

Generated by LIRA (Learning Intrusion Reasoning Agent). This report supports security triage and does not replace human verification.